

MOST REPEATED EXAM QUESTIONS

Comprehensive Analysis of Past Papers (2023–2024)

Course: T.E. Computer Engineering (Semester V) • Subject: Graph Theory (TE1158 / CE531)

Executive Summary: This analytical document identifies the highest-frequency question patterns from five consecutive exam cycles: *January 2023, June 2023, December 2023, May 2024, and December 2024*. Questions are sorted by their recurrence frequency. Mastering these top 10 core concepts covers approximately **75–80%** of the total marks allocated in any standard variant of this paper.

Core High-Frequency Question Matrix

1. The Hungarian Matching Algorithm

5 / 5 Papers (100% Recurrence)

Appeared in: Jan 2023 (Q4a), June 2023 (Q4a), Dec 2023 (Q4a), May 2024 (Q4a), Dec 2024 (Implicit matrix variants)

This question is repeated almost exactly word-for-word across multiple years, using an initial matching scenario with a bipartite graph structure of 5 workers/people and 5 or 10 jobs.

Sample Exam Wording (May 2024 / June 2023):

In a company $X = \{x_1, x_2, x_3, x_4, x_5\}$ people have applied for $Y = \{y_1, y_2, y_3, y_4, y_5\}$ jobs. One person can qualify for more than one job. Following graph depicts person-job qualification details. Demonstrate the Hungarian Algorithm to obtain a job match for every person (Initial matching M is indicated by the heavy edges). [8 Marks]

2. Fundamental Proofs on Tree Properties

5 / 5 Papers (100% Recurrence)

Appeared in: Jan 2023 (Q2A), June 2023 (Q3c), Dec 2023 (Q3), May 2024 (Q3c), Dec 2024 (Q1b, Q8b)

Every single paper requires proving one or more of the fundamental structural properties of structural trees. The most commonly rotated sub-proofs include:

- Prove that a tree with n vertices has exactly $n-1$ edges (Dec 2024, Jan 2023).
- Prove that in a tree, any two vertices are connected by a unique path (Jan 2023, June 2023, May 2024).
- Prove that every edge in a tree is a cut edge / bridge (June 2023, May 2024).
- Prove that every tree is a bipartite graph and identify which trees are complete bipartite graphs (Dec 2023, Dec 2024).

3. Bipartite Graph Cycle Characterization Theorem

4 / 5 Papers (80% Recurrence)

Appeared in: Dec 2023 (Q1d), June 2023 (Q2a), May 2024 (Q2a), Dec 2024 (Q7a)

A benchmark theoretical question asking to establish the characterization of bipartite structures based on the lengths of its cycles.

Standard Exam Wording:

Define a bipartite graph with the help of an example. Prove that a simple graph G is bipartite if and only if it contains no odd length cycles (or each cycle of G has even length). [5–6 Marks]

4. Fleury's Algorithm for Eulerian Trails

4 / 5 Papers (80% Recurrence)

Appeared in: Jan 2023 (Q2B), June 2023 (Q2b), Dec 2023 (Q2b), May 2024 (Q2b), Dec 2024 (Q2c)

Students are given a specific undirected network graph and required to systematically trace out a valid Eulerian trail.

Standard Exam Wording:

Explain Fleury's algorithm to produce an Eulerian trail. Apply it to the given graph to trace out a clear Eulerian line sequence. [6–7 Marks]

5. Graph Coloring Practical Optimization Modeling

4 / 5 Papers (80% Recurrence)

Appeared in: Jan 2023 (Q5C), June 2023 (Q5c/Q5a), May 2024 (Q5a/Q6c), Dec 2024 (Q4b/Q7d)

This question appears as a word problem requiring translation into a vertex coloring optimization task. The common context variants are:

- **Lecture Timetable Scheduling:** Given an asterisk collision table showing which classes conflict, find the minimum time slots needed (Jan 2023, June 2023).
- **Chemical Warehouse Storage:** Given a list of reactive chemicals that cannot be stored adjacent to one another, find the minimum storage sections required (May 2024, Dec 2024).
- **Sudoku Puzzle Solver:** Model and demonstrate a 4×4 Sudoku grid solution using independent color sets (June 2023, May 2024).

6. Handshaking Theorem & Dilemma Verification

4 / 5 Papers (80% Recurrence)

Appeared in: Jan 2023 (Q1C/Q5A), June 2023 (Q4b), Dec 2023 (Q5a), Dec 2024 (Q3a)

Asks for state/proof formatting of the theorem, often coupled with verifying it on an accompanying diagram or tournament digraph.

Standard Exam Wording:

State and prove the Handshaking Theorem for an undirected graph. Alternatively: State the Handshaking dilemma on a directed graph and verify the relation for the following tournament diagram. [5–8 Marks]

7. Tournament Score Sequences & Sub-cycles

4 / 5 Papers (80% Recurrence)

Appeared in: Jan 2023 (Q8D), June 2023 (Q8d), Dec 2023 (Q2c), May 2024 (Q8c/Q6b)

Focuses on analyzing a tournament's directed components to determine out-degree scores and identify standard cyclic sizes.

Standard Exam Wording:

The score of a vertex of a tournament is its out-degree, and the score sequence is formed by arranging the scores in non-decreasing order. Obtain the score sequence for the given tournament. In the same tournament, find sub-cycles of length 3, 4, and 5. [5–6 Marks]

8. Core Definitions: Structural Cuts & Paths

3 / 5 Papers (60% Recurrence)

Appeared in: Jan 2023 (Q2C/Q8B), June 2023 (Q1c/Q2c), Dec 2023 (Q1c/Q2a)

A definition collection question requiring crisp explanations and clean custom illustrative diagrams.

Standard Exam Wording:

Define the following terms and provide suitable clear examples for each: (i) Bridge/Cut-edge, (ii) Disconnecting Set, (iii) Cut Vertices, (iv) Walk, (v) Trail, (vi) Path, (vii) Clique. [5–8 Marks]

9. Floyd-Warshall All-Pairs Shortest Path Algorithm

3 / 5 Papers (60% Recurrence)

Appeared in: Jan 2023 (Q7B), June 2023 (Q3a), Dec 2024 (Q3c)

A rigorous computation numerical problem requiring setting up distance matrices sequentially.

Standard Exam Wording:

Apply the Floyd-Warshall algorithm to find the all-pairs shortest paths or distances between every pair of vertices for the given weighted directed/undirected graph network. Show the state of matrices explicitly. [5–6 Marks]

10. Graph Isomorphism Verification Matrix / Bijection

3 / 5 Papers (60% Recurrence)

Appeared in: Jan 2023 (Q1B), May 2024 (Q1b), Dec 2024 (Q2b/Q3b)

Testing structural parity between two provided topologies.

Standard Exam Wording:

Define Graph Isomorphism. For the pair of graphs below, decide if they are isomorphic or not. If they are isomorphic, give the vertex bijection (or show via adjacency matrices). If they are not, justify using invariant properties. [5–6 Marks]